

WELCOME TO

# OpenClaw MM Rubrics

BLUE SHELL

Contributor Course · June 2026

Build grounded, multimodal evaluation tasks where the initial agent trajectory **fails**  
 **$\geq 50\%$  of rubric weight** — then document, guide, and grade.

# Blue Shell

## Course Overview

The goal of this course is to ensure that everyone contributing to the OpenClaw MM Rubrics Blue Shell project has a strong and consistent understanding of the project's core requirements, workflows, and quality standards.

Work through each section in order. After every section, answer the **Knowledge Check** questions. At the end, complete the **Final Screening**; passing requires demonstrating you can design grounded, difficult, gradable tasks — not just memorize definitions.

## Project Resources

- [🔗 Project Attempter Guidelines](#) → Complete guidance on how to plan, execute, and evaluate tasks in the OpenClaw MM Rubrics – Blue Shell project.
- [🔗 Outlier Community Project Welcome Page](#) → The central hub for the project. Use this space to stay informed on project updates, ask questions, share insights, and connect with other team members.

🗨️ *Please note that changes within the project are part of the day to day workflow. Because of that, we strongly recommend always keeping the project guidelines handy, as well as regularly reviewing the project channel in the Outlier Community, since all important updates and changes are announced there.* 🗨️

## ⚠️ Important notes

- This course is not graded. However, if you miss the concepts covered here, it will be extremely difficult to pass the upcoming screener.

## Intro

## SECTION 1

# Introduction

What you're building and why

- Design **single-turn** tasks that test OpenClaw agents across tools, media, and reasoning
- Goal is **not** first-try success — initial trajectory must **fail**  $\geq 50\%$  of total rubric weight
- Then: spot real failures → silver trajectory to correct answer → rubrics ranked against **initial** run

| STEP | NAME              | GOAL                                                      |
|------|-------------------|-----------------------------------------------------------|
| 0    | Initial Idea      | Explore universe first; align with Category & Subcategory |
| 1    | Story Draft       | Agent Objective · MM zip · Desired Outcome                |
| 2    | Run Prompt        | Final prompt → OpenClaw (one interaction)                 |
| 3    | Spot Mistakes     | Identify every genuine model failure                      |
| 4    | Silver Trajectory | Guide agent to correct solution                           |
| 5    | Rubric Generation | Create rubrics · Present/Not Present · justify failures   |



# Blue Shell

You are building **evaluation tasks** for OpenClaw agents operating in realistic personal and professional universes. Each task tests whether a model can independently plan, coordinate tools, reason across multimodal evidence, and produce verifiable deliverables.

Your job is not to make the agent succeed on the first try. It is to design a **realistic, grounded, sufficiently difficult** scenario where the initial trajectory **fails at least 50% of the final rubric score**, then document those failures precisely and guide a silver trajectory to the correct solution.

## What Makes a High-Quality Task

A deliverable task package includes:

| Component              | Purpose                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Universe-grounded idea | Scenario solvable only with tools/data in Universe Explorer                                                                      |
| Story draft            | Agent Objective, MM zip upload, Desired Outcome                                                                                  |
| Prompt                 | Single-turn, natural-language request aligned with the Universe, Category, and Subcategory — <b>after exploring the universe</b> |
| Initial trajectory     | One interaction; agent fails $\geq 50\%$ of rubric weight                                                                        |
| Silver trajectory      | Follow-up guidance until the correct solution is found                                                                           |

## What Makes a High-Quality Task

A deliverable task package includes:

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| Story draft            | Agent Objective, MM zip upload, Desired Outcome                                                                                  |
| Prompt                 | Single-turn, natural-language request aligned with the Universe, Category, and Subcategory — <b>after exploring the universe</b> |
| Initial trajectory     | One interaction; agent fails $\geq 50\%$ of rubric weight                                                                        |
| Silver trajectory      | Follow-up guidance until the correct solution is found                                                                           |
| Rubrics + rankings     | Binary pass/fail per criterion against <b>initial</b> trajectory                                                                 |
| Failure justifications | For every failed rubric on the initial run                                                                                       |

## Task Parameters (Non-Negotiable)

- **Single-turn only** — entire request in one self-contained prompt; no iterative user guidance in the initial run

## Project Overview

### SECTION 2

# Project Overview

Core dimensions every task must satisfy

#### THREE STAGES

- **Acquire** — tools, files, media
- **Process** — reconcile, rank, score
- **Output** — verifiable artifact

#### MUST INCLUDE

- Multi-system coordination
- Real decision logic (not just tool calls)
- Realistic friction & messy data
- MM required for core work

- **Blind Test Rule:** score without media must be <50% of score with media
- **Single-turn only** · solvable only via Universe Explorer loadout
- **pass@8 ~40%** for top models — 100% fail = broken rubric/task

**NOT acceptable:** decorative images · text-only solvable · single-step workflows · impossible given loadout



# Blue Shell

OpenClaw agents must demonstrate **multi-system coordination** across at least three stages:

- 1. **Data Acquisition** — gather evidence from tools, files, and media
- 2. **Processing / Reasoning** — reconcile, rank, score, or apply rules
- 3. **Output Generation** — produce a defined, verifiable artifact

## Core Dimensions (Every Task Must Score Here)

| Dimension               | What It Measures                             | Expectation                                                 |
|-------------------------|----------------------------------------------|-------------------------------------------------------------|
| Complex                 | Branching, recovery, tool/source diversity   | Meaningful planning across heterogeneous artifacts          |
| Long-horizon            | Trajectory length, persistent state          | Multi-step workflows; MM artifacts require real inspection  |
| Objective               | Gradable outputs from stated rules/sources   | Clear formats and verifiable sources of truth               |
| Multimodal              | Visual/audio/media required for core work    | Removing media should cripple performance (Blind Test Rule) |
| Cross-modal / cross-API | Evidence combined across modalities/services | Outputs from one step feed the next                         |
| Initial failure         | Task difficulty                              | ≥50% of total rubric weight fails on initial trajectory     |

### Architectural Requirements (All Mandatory)

Your task must include **all** of the following. If it can be completed in a short linear interaction, it is too simple.

- **Multi-stage coordination** — Acquire → Process → Decide → Output with dependencies
- **Real decision logic** — ranking, scoring, thresholds, rule-based comparisons (not just tool chaining)
- **Multi-system coordination** — multiple servers/sources that cannot be replaced with text-only reasoning
- **Realistic friction** — messy data, ambiguity, conflicting sources, missing fields
- **Model differentiation** — models should not all produce identical output
- **Defined artifact** — clear, verifiable outcome (not exploratory chat)
- **Structural reasoning** — logical sequencing and persistent state when relevant

### NOT Acceptable

- Single-step workflows, simple summarization, basic search
- Static reasoning without tool coordination
- Tasks impossible given the universe loadout
- Tasks likely to produce identical performance across models
- Decorative multimodal inputs (solvable without inspecting media)

# Blue Shell

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- Tasks likely to produce identical performance across models
- Decorative multimodal inputs (solvable without inspecting media)

## Categories & Subcategories

| Category            | Subcategories                                                                           |
|---------------------|-----------------------------------------------------------------------------------------|
| Visual Learning     | Homework/Problem Solving · Lab/Fieldwork Documentation · Textbook/Lecture Comprehension |
| Commerce & Product  | Visual Shopping/Comparison · Product Listing QA · Brand/Packaging Audit                 |
| Creative & Media    | Image/Video Editing · Social Media Content Audit · Design/Portfolio Review              |
| Operations & QA     | Document/Receipt Processing · Inventory Visual Audit · UI/UX Screenshot Audit           |
| Health and Wellness | Skin/Symptom Triage · Nutrition/Meal Logging                                            |
| Property and Space  | Real Estate Listing Review · Interior Design/Renovation                                 |

## Building the Idea

### STEP 0

#### SECTION 3

## Building the Idea

Universe first — not the prompt

- Open **Universe Explorer** before drafting: servers, data, calendar dates, contacts
- Add **Environment ID + Service Universe Artifact ID** (required)
- Prompt must be solvable using **only** what appears in the loadout
- Use SQL/DB explorer for data not visible in the UI

#### DEFINE AT STEP 0

- Real persona & stakes
- Problem to solve
- Which servers + MM inputs
- Planned failure levers

#### COMMON AUDIT FAILS

- Feasibility with tools
- Trivial integration (image→CSV only)
- Contrived / too-clean inputs

Gmail when only **messaging** exists = environment defect, not model failure



### Start With the Universe — Not the Prompt

Before drafting anything, open **Universe Explorer** for your assigned task and investigate:

1. Which **servers/tools** are in the loadout (email, fintrack, zillow, fitbit, etc.)
2. What **data** exists (calendar dates, contacts, transactions, messages)
3. What **constraints** are realistic for this persona
4. Whether your scenario requires servers that **actually exist** in this universe

*Critical rule: The prompt must be solvable using only what Universe Explorer shows. Designing for Gmail when only messaging exists is an environment defect, not a model failure.*

STEP 0

SECTION 3

## Universe Reconnaissance Checklist

Verify the universe before writing the prompt

- ▶ Add **Environment ID + Service Universe Artifact ID**
- ▶ Confirm **required servers** (e.g., fintrack + calendar + image)
- ▶ Check **calendar date ranges** — universe calendars are static

#### Artifact ID Values

| Artifact                     | Artifact ID    |
|------------------------------|----------------|
| Environment ID               | ENV-7F3A-21B4  |
| Service Universe Artifact ID | SVCU-9C2D-8E11 |

## STEP 0

## SECTION 3

# Universe Reconnaissance Checklist

Verify the universe before writing the prompt

- ▶ Add **Environment ID + Service Universe Artifact ID**
- ▶ Confirm **required servers** (e.g., fintrack + calendar + image)
- ▶ Check **calendar date ranges** — universe calendars are static
- ▶ Use **SQL / DB explorer** for data not visible in the UI
- ▶ Verify persona-appropriate **category/subcategory fit**

## KEY TOOLS

- ▶ Universe Creator
- ▶ SQL / DB Explorer
- ▶ Agent Chat (persona)
- ▶ JSON / CSV / XML export

## WATCH OUT FOR

- ▶ Static calendars — may be empty
- ▶ Missing servers = defect
- ▶ Category mismatch breaks grading

Static calendar dates with no data = environment setup gap, not a model failure

### Artifact ID Values

| Artifact                     | Artifact ID    |
|------------------------------|----------------|
| Environment ID               | ENV-7F3A-21B4  |
| Service Universe Artifact ID | SVCU-9C2D-8E11 |
| Calendar Artifact ID         | CAL-4B7E-6D02  |

### Paste Artifact ID

Enter the Service Universe Artifact ID

Create Universe

## Drafting the Idea

At this step, you are **not** writing the final prompt. You are defining:

- **Stepping into the Universe user's shoes** (real persona with real stakes)
- **What problem** do they need solved
- **Which universe systems** must be involved
- **Which multimodal inputs** will be necessary (not decorative)
- **What failure levers** naturally fit

## Idea Quality Signals

| Strong Idea                                          | Weak Idea                    |
|------------------------------------------------------|------------------------------|
| Grounded in actual universe data                     | Generic assistant request    |
| Requires cross-server + MM reconciliation            | Single-source lookup         |
| Natural friction (messy photos, conflicting records) | Clean textbook dataset       |
| Clear deliverable filename(s) planned                | Vague "give me a report."    |
| Category/subcategory fit is obvious                  | Forced or unrelated scenario |

## Blue Shell

|                                                      |                              |
|------------------------------------------------------|------------------------------|
| Natural friction (messy photos, conflicting records) | Clean textbook dataset       |
| Clear deliverable filename(s) planned                | Vague "give me a report."    |
| Category/subcategory fit is obvious                  | Forced or unrelated scenario |

## Common Errors

| Issue                  | Fix                                                                    |
|------------------------|------------------------------------------------------------------------|
| Feasibility with tools | Re-explore the universe; redesign around available servers             |
| Trivial integration    | Add cross-server handoff (value from image → lookup in fintrack)       |
| Category relevance     | Re-read subcategory definition; adjust scenario                        |
| Contrived inputs       | Use messy real-world files, not curated perfect JPGs                   |
| Unrealistic scenarios  | Think like the user — use the universe, context, and expected queries. |

## Story Draft

## STEP 1

## SECTION 4

# Story Draft

Agent Objective · MM Upload · Desired Outcome

## AGENT OBJECTIVE

- Real persona, real problem, high-level success
- **No** "Act as X agent" or tool lists
- Describe **what** success looks like, not how
- **MM zip**: downsample (480p video ·  $\leq$ 1080p images · mp3 64kbps) · messy realistic files
- **$\leq$ 20% LLM-generated** · never LLM .xlsx / .docx / .pdf · no HEIC · no answer-leak filenames
- Story draft is **not embedded** in OpenClaw — you guide the agent via prompts in trajectories

## DESIRED OUTCOME

- Artifact structure, columns, enums, logic
- **Shape** of success — not answer key
- Never leak tool-fetched values or Q1:B answers



Format rules for rubrics must also appear in the **prompt** — Desired Outcome alone is invisible to the model

## The Story Draft Is Your Foundation

If the story draft is rushed, the agent will succeed too easily, and you cannot meet the **≥50% rubric failure** requirement.

*Remember: The story draft (Agent Objective, Desired Outcome, MM upload) is not embedded in OpenClaw. Through each trajectory, you must guide the agent with prompts to produce what you defined.*

### Agent Objective

**Purpose:** Define the real-world persona, concrete problem, context, and high-level success — **without implementation steps.**

| Include                                  | Avoid                               |
|------------------------------------------|-------------------------------------|
| Real persona and stakes                  | "Act as X agent" role-play          |
| What success looks like (artifact types) | Tool-by-tool instructions           |
| Constraints as outcomes                  | Policy thresholds leaked from files |
| Why the task matters                     | Numbered implementation pipeline    |

**Good pattern:** "Jordan owns an online running shoe store and needs to triage five customer complaint emails with photos before replying today."

**Bad pattern:** "Act as the escalation agent; use Gmail, Supabase, and the uploaded images to..."

## Blue Shell

### Multimodal Upload (Zip File)

#### Required

- **Zip** file with all context needed
- **Downsampled** files before upload (videos → 480p, images → ≤1080p, audio → mp3 64kbps)
- **Realistic messiness:** blurry shots but still visible, duplicates, neutral filenames, mixed orientations

#### Strict Rules

| Rule                   | Detail                                                           |
|------------------------|------------------------------------------------------------------|
| LLM-generated cap      | ≤20% of total artifacts                                          |
| Restricted LLM formats | Never LLM-generate .xlsx, .docx, .pdf                            |
| No answer leaks        | Filenames, manifests, and helper docs must not reveal answers    |
| No junk files          | No .env, .key, meaningless junk01.txt patterns                   |
| No HEIC                | Convert to JPG/PNG                                               |
| Copyright              | CC0/CC-BY OK; avoid CC-BY-NC, Unsplash, Getty, paywalled sources |
| PII                    | Avoid faces of non-public figures; use synthetic persona data    |

## Blue Shell

|           |                                                                  |
|-----------|------------------------------------------------------------------|
| Copyright | CC0/CC-BY OK; avoid CC-BY-NC, Unsplash, Getty, paywalled sources |
| PII       | Avoid faces of non-public figures; use synthetic persona data    |

**Realism Standard:** A folder of perfectly cropped JPGs with clean names is *contrived*. Real users upload IMG\_0427.jpg, skewed scans, and mismatched filenames.

## Desired Outcome

Defines the **inspectable end state** — not a copy of the Agent Objective.

| Include                                | Avoid                                                 |
|----------------------------------------|-------------------------------------------------------|
| Required artifact names and structure  | Verbatim answer key (e.g., "Q1: B, Q2: B")            |
| Column names, enums, decision logic    | Tool-fetched values leaked (balances, names from API) |
| How verification will work             | Vague "good quality output."                          |
| Conditional branches (which file when) | Subjective criteria without anchors                   |

**The more detail, the better** — but detail should describe **shape**, not **pre-solved content**.

## Blue Shell

| Include                                | Avoid                                                 |
|----------------------------------------|-------------------------------------------------------|
| Required artifact names and structure  | Verbatim answer key (e.g., "Q1: B, Q2: B")            |
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| How verification will work             | Vague "good quality output."                          |
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**The more detail, the better** — but detail should describe **shape**, not **pre-solved content**.

### Desired Outcome vs. Prompt

- **Desired Outcome** = your internal ground truth for rubric authoring
- **Prompt** = what the model actually sees
- Any format requirement you want to **enforce in rubrics** must **naturally** appear in the **prompt**, not only the Desired Outcome

## Prompt Generation

STEP 2

SECTION 5

# Prompt Generation

Single-turn · realistic voice · aligned with story draft

- Name **every output filename** in the prompt (top audit fail if missing)
- Policies & thresholds live in **input files** — not hard-coded in prompt
- Natural user voice · real call to action · genuine MM reasoning required
- Design around **failure levers**: cross-source conflict · skipped conditionals · filename traps · handwriting OCR · implicit lookups

### AVOID

- Step 1, Step 2 checklists
- Answer leaks & decorative MM
- Long audio (>5–10 min)
- "Asking for the sake of asking"

### NOT MODEL FAILURES

- prompt-error / timeout
- HEIC crashes
- Missing universe data
- Format only in Desired Outcome



## Blue Shell

The prompt is a **single-turn**, realistic user message — fully aligned with Agent Objective and Desired Outcome.

### Prompt Checklist

- Sounds like a real person (not "You are the X agent")
- Names **every required output filename** in the prompt body
- Requires genuine multimodal reasoning
- Policies/thresholds live in **input files**, not hard-coded in prompt
- Natural constraints embedded in user voice
- Closes with a real call to action
- Not solvable through text-only reasoning
- Minimum grammatical consistency

| Avoid                               | Why                                                     |
|-------------------------------------|---------------------------------------------------------|
| Checklist-style "Step 1, Step 2..." | Reads like developer spec, not user request             |
| Hidden answer leaks / obvious clues | Invalidates evaluation                                  |
| Infinite unrelated requests         | Artificial complexity ("asking for the sake of asking") |

## Blue Shell

- minimum grammatical consistency

| Avoid                                | Why                                                           |
|--------------------------------------|---------------------------------------------------------------|
| Checklist-style "Step 1, Step 2..."  | Reads like developer spec, not user request                   |
| Hidden answer leaks / obvious clues  | Invalidates evaluation                                        |
| Infinite unrelated requests          | Artificial complexity ("asking for the sake of asking")       |
| Missing output filenames             | <b>#1 audit fail code</b> — rubrics cannot grade consistently |
| Format rules only in Desired Outcome | The model never sees them — a rubric defect if enforced       |
| Long audio (>5–10 min)               | Causes timeout / prompt-error failures                        |

## Output Filename Precision

Tasks often fail when:

- MEMORY.md is referenced as if it has to be created from scratch despite it lives in the Agent Enviroment.
- The prompt asks for property\_match\_review.md, but rubrics grade gabriela\_listing\_audit.csv
- MEMORY.md or auxiliary files omitted from the prompt but required in rubrics

**Rule:** Every file your rubric grades must be named in the prompt with unambiguous creation/update intent.

Model Failure Levers (Design Prompts Around These)

Use weaknesses from real trajectories — not environment bugs.

| Lever                        | Example Scenario                                                                            |
|------------------------------|---------------------------------------------------------------------------------------------|
| Single-source reliance       | Notebook says skip watering Day 5; fridge log shows water — model reports watered every day |
| Fabricating precise numbers  | No ruler in photos; model invents 0.5, 1.5, 2.5 cm measurements                             |
| Filename-as-evidence         | parking.jpg shows an oil stain; the model writes, "pothole confirmed."                      |
| Handwritten digit confusion  | Messy worksheet; 6 vs 8 cascades through grading                                            |
| Skipped conditional          | "Draft email IF dinner > budget" — dinner under budget; model drafts anyway                 |
| First-text-match wins        | recipes.txt lists Vegetable Soup first; the video shows chicken                             |
| Skill not discovered         | Email available; model uses messaging                                                       |
| Auto-correcting user content | "Preserve my kid's spelling" — model normalizes to standard English                         |
| Missed implicit prerequisite | "Use my liquidity as base" — model estimates instead of querying fintrack                   |

## Blue Shell

|                              |                                                                             |
|------------------------------|-----------------------------------------------------------------------------|
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## What Is NOT a Model Failure

Do not count these toward the 50% threshold:

- openclaw:prompt-error, idle timeout, session abort
- HEIC / image tool dependency crashes
- Unsolvable task (missing universe data)
- The rubric demands a format that was never stated in the prompt
- Environment is missing the required server

STEPS 3 & 4

### SECTION 6

# OpenClaw Workflow

Model trajectory · Silver trajectory

#### INITIAL TRAJECTORY

- One prompt → one response chain
- Close tab · don't force session end
- **Collect Traces & Continue**
- Confirm **≥50% rubric weight** fails for real reasons
- If too easy → add cross-modal / conditional complexity
- Document: **what happened → which lever → which rubric catches it**
- Silver must show cross-modal reconciliation · ≥3 stages · branching/recovery

#### SILVER TRAJECTORY

- Continue from original interaction
- **Restore to Seed** — never Start Fresh
- Follow-up prompts until correct solution
- Proves task is solvable with loadout



Rubrics grade the **initial** trajectory — silver proves solvability, not rubric rankings

## Execution: Initial (Model) Trajectory

### Procedure

1. Upload zip or individual MM files
2. Open OpenClaw in a new tab
3. Send **only** the prompt (single interaction for trace collection)
4. Close the tab **without** forcing the session end
5. Use **Extract Trajectories / Collect Traces & Continue**
6. Download trajectories and workspace when complete

### Critical Reminders

- **Exactly ONE trajectory and ONE workspace** from the initial interaction
- If you keep iterating in the same trace, **reset and rerun** — the original trace must be one prompt → one response chain
- Do **not** deliver failed/incomplete trajectories

### The 50% Failure Gate

After collecting the initial trajectory:

1. List **every genuine model mistake**
2. Estimate the rubric weight that would fail
3. If <50%, **increase task difficulty** — add cross-modal reconciliation, implicit cross-check, conditional logic
4. Do **not** pad with format micro-spec or unsolvable rubrics

## Spot Trajectory Mistakes

For each mistake, ask:

1. Was correct information available?
2. Did the model make a **decision** (not an environment crash)?
3. Would a rubric criterion capture this failure?
4. Is this a **single** mistake (not 20 per-row rubric explosions)?

Document: **what happened** → **which lever** → **which rubric will catch it**.

## Silver Trajectory

### Purpose

Prove the task is **solvable** with the loadout and distinguish:

- Real model failures
- Environment crashes
- Rubric defects

# Rules

| Do                                                   | Don't                                                           |
|------------------------------------------------------|-----------------------------------------------------------------|
| Continue from the original interaction               | Use <b>Start Fresh</b> (deletes initial trajectory permanently) |
| Use <b>Restore to Seed</b> to preserve initial work  | Collect traces while iterating in the initial session           |
| Add follow-up prompts correcting specific errors     | Assume story draft fields are visible to the agent              |
| Use override model calls if the agent never succeeds | Advance without all positive rubrics satisfied                  |
| Select (Re-) Collect Traces & Continue when done     |                                                                 |

## Silver Quality Bar

Silver trajectory must show:

- Cross-modal reconciliation (media + at least one other source)
- Non-trivial handoff between sources
- ≥3 stages (Acquire → Process → Output)
- Branching or recovery behavior
- Solvable using **only** the universe loadout

### STEP 5

#### SECTION 7

# Rubric Building

Create · rank · justify

- **Self-contained · atomic · outcome-based · positive phrasing** (weight = polarity)
- Weights = **difficulty to verify** (+5/3/1 · -5/3/1) — NOT importance
- Negatives  $\leq 30\%$  · only for observed likely deviations
- Prompt & rubrics must grade **same deliverables** — no oracle hard-coding

#### RATINGS (INITIAL RUN)

- Positive + Not Present = **fail**
- Negative + Present = **fail**
- Failed rubrics need **3 justifications**

#### TOP DEFECTS TO AVOID

- Deliverable mismatch (P0)
- Over-specification / bundling
- Incorrect Present/Not Present
- Input-perception rubrics

Can't justify all 3 areas? **Delete the rubric.** Use spot checks for repeated actions — not 20 per-row criteria.



Each criterion is a **binary** pass/fail statement about whether the initial agent met a requirement.

## Ratings

| Weight Sign         | Not Present                                      | Present                                      |
|---------------------|--------------------------------------------------|----------------------------------------------|
| Positive (+1/+3/+5) | <b>Failure</b> — the model should have done this | Success                                      |
| Negative (-1/-3/-5) | Success — bad behavior absent                    | <b>Failure</b> — the model did the bad thing |

**|** *Failed rubrics (positive Not Present OR negative Present) require **three-part justifications**.*

## Rubric Fundamentals

### Core Rules

1. **Outcome-based** — grade deliverables and state changes, not "agent used tool X" unless intent requires it
2. **Self-contained** — evaluable from model response/trajectory alone; include specific values in the criterion text, especially when the rubric needs the output image to judge.
3. **Atomic** — one evaluable idea per criterion (schema-check criteria are an allowed exception)
4. **Positive phrasing** — even negatives are written as positive statements; the weight sign carries polarity
5. **Prompt-aligned** — every literal string (filename, heading, value) must appear in the prompt or be discoverable without oracle leaks
6. **Intent over literal format** — grade whether the user request was satisfied, not the exact phrasing the prompt never required

## Self-Containment Example

× "Response identifies the first president of the USA." ✓ "Response identifies the first president of the USA as George Washington."  
× "xyz.csv includes the first row from abc.png" (requires opening input) ✓ "xyz.csv includes the first non-header row 'John Smith', '123-456-7890', '\$300'"

**Note:** Input file contents are external for self-containment. Criteria requiring input+output comparison should embed the expected value or use deterministic spot checks.

## Rubric Categories

- Task Completion
- Instruction Following
- Factuality and Hallucination
- Tool Use
- Agent Behavior
- Safety & Boundaries (when applicable)

## Evaluation Targets

| Target                | Use For                                                               |
|-----------------------|-----------------------------------------------------------------------|
| final answer artifact | Files created/modified in workspace                                   |
| state change          | External service mutations (email sent, calendar event created)       |
| user-facing message   | What assistant tells user in the final answer of the chat             |
| trajectory            | Required reasoning path/scope discipline (not exact tool call counts) |

## Repeated Actions → Spot Checks

Do not create 20 per-row rubrics for one mistake.

- **Aggregate count:** *"Agent sends all 16 required emails."*
- **Spot instances:** *"Emails #2, #7, and #14 contain correct personalized information"*
- Max ~5 spot checks per group; add volume criterion

## Stacked Rubrics (OR Logic)

Use when multiple outcomes are **genuinely equivalent** — not to patch missing coverage.

# Weights: Difficulty-Based

Weights reflect how hard the criterion is to verify — NOT how important it feels.

| Weight | Difficulty         | Heuristic                                                                    |
|--------|--------------------|------------------------------------------------------------------------------|
| +5     | High               | ≥3 difficulty axes (tool coordination, reasoning depth, modality, discovery) |
| +3     | Medium             | Exactly 2 axes                                                               |
| +1     | Low                | 0–1 axes; mechanical single-source check                                     |
| -5     | Common failure     | Model strongly tempted (filename trust, wrong default tool)                  |
| -3     | Moderate avoidance | Plausible wrong default with some signal to avoid                            |
| -1     | Rare failure       | The model rarely does this unprompted                                        |

## Difficulty Axes

- **Tool/source coordination** — 3+ sources reconciled or tools chained
- **Reasoning depth** — multi-step inference, conditionals, calculations
- **Modality** — cross-modal (image + structured data + text)
- **Discovery effort** — SQL, hidden fields, cross-document tracing

## Weight Examples

| Criterion                                                                       | Weight | Why                     |
|---------------------------------------------------------------------------------|--------|-------------------------|
| Reconcile the receipt image against Fintrack AND apply policy PDF rules         | +5     | 3+ axes                 |
| Extract value from the image and report correctly                               | +3     | Single cross-modal step |
| Confirm <code>audit_report.md</code> exists                                     | +1     | Mechanical              |
| Trusts <code>parking.jpg</code> filename without inspecting the oil-stain image | -5     | Strong temptation       |

## Negative Rubrics

- **Target:** ~25% of criteria negative
- **Hard ceiling:** 30%
- Prefer **positive** framing of requirements first
- Add negatives only for **observed** likely deviations in the initial trajectory

## Rubric Justifications

Required for every failed rubric. If you cannot complete all three, **delete the rubric**.

1. **Why the rubric is correct** — Quote prompt/input policy that grounds the check
2. **Where the model made a mistake** — Cite specific trajectory turn/tool/output
3. **Why necessary for a correct answer** — Explain the impact on task success

## Justification Rules

- Definitive, not speculative
- Reference prompt or trajectory directly
- Never cite other models or pass rates
- Unsure? Delete the criterion

## Pre-Submit Rubric Checklist

- Prompt and rubrics grade the **same deliverables**
- Every literal string in rubrics appears in the prompt (or the policy file referenced in the prompt)
- Weights reflect **difficulty**, not importance
- Negatives  $\leq 30\%$ ; positives cover all prompt requirements first
- No contradictory criteria
- Spot checks + volume for repeated actions
- Rankings reflect **initial** trajectory only
- Failed rubrics have 3-part justifications
- Initial trajectory fails  $\geq 50\%$  of total rubric weight

# Common Errors

| Defect                             | Prevention                                                                    |
|------------------------------------|-------------------------------------------------------------------------------|
| Prompt/rubric deliverable mismatch | Same filenames in the prompt and rubrics                                      |
| Over-specification                 | Grade intent; allow equivalent phrasings                                      |
| Rubrics test unstated requirements | Trace every criterion to the prompt text                                      |
| Hard-coded oracle values           | Use live-tool values; embed in a self-contained criterion from the actual run |
| Contradictory rubrics              | Check positive/negative polarity pairs                                        |
| Non-atomic bundling                | Split independent assertions                                                  |
| Incorrect Present/Not Present      | Re-read the trajectory evidence before rating                                 |
| Input perception rubrics           | Prefer output-checkable criteria; embed specifics                             |
| Multi-file visual checks           | One file per criterion or deterministic tests                                 |
| Weak/missing justifications        | All 3 justification areas required                                            |

## SECTION 8

# You're Ready

Final quality gates before submit

| GATE              | ASK YOURSELF                                                                |
|-------------------|-----------------------------------------------------------------------------|
| <b>Grounded</b>   | Did I explore the universe and use only available servers/data?             |
| <b>Multimodal</b> | Would the task collapse without uploaded media?                             |
| <b>Difficult</b>  | Does initial trajectory fail $\geq 50\%$ of rubric weight for real reasons? |
| <b>Solvable</b>   | Does silver trajectory succeed with the loadout?                            |
| <b>Gradable</b>   | Do rubrics align with prompt, self-contain, use difficulty weights?         |
| <b>Documented</b> | Are all failed rubrics justified with prompt + trajectory evidence?         |

**Quick ref:** Universe → Idea → Story → Prompt → Initial ( $\geq 50\%$  fail) → Silver → Rubrics (initial) → Justify



You are shipping a **training signal** for frontier agents. A weak task teaches nothing; a strong task cleanly separates capable reasoning from predictable failure modes.

## When to Escalate

Ask for help when:

- Universe Explorer shows that your category requires missing servers
- Initial trajectory fails for environment errors, not model decisions
- You cannot reach 50% failure without adding artificial requirements
- Rubric evaluator flags >10% major errors, and you cannot resolve them

## You Are Ready When You Can

1. Explain why your universe **requires** specific servers for your scenario
2. Point to three **failure levers** in your prompt design
3. Defend every rubric criterion against the **prompt text**
4. Show where the initial trajectory failed and why it was the model's fault
5. Demonstrate silver trajectory success without changing the task premise

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5. Demonstrate silver trajectory success without changing the task premise

The most important thing to remember: **quality over speed**. A rushed Story Draft, a poorly grounded prompt, or bundled rubrics will fail quality review and require complete rework. Take the time to explore your universe, use the checklist before submitting, and when in doubt — split the rubric criterion, quote the prompt, and delete what you can't justify.

Good luck on the screening, and welcome to the project 🦀